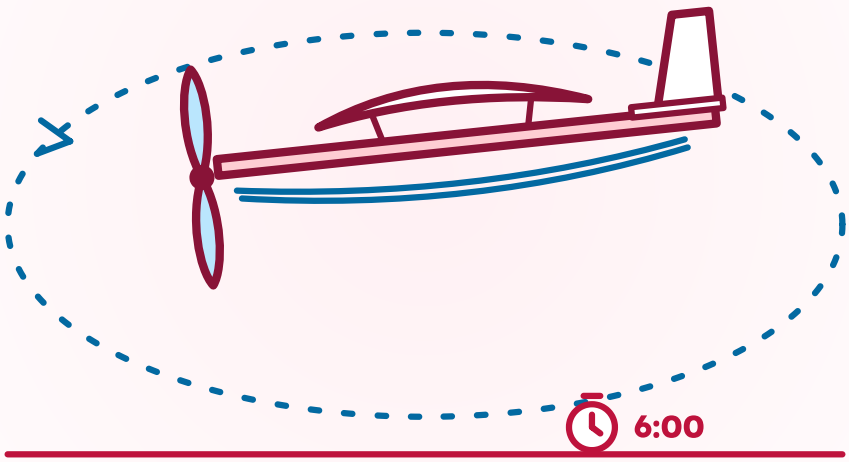
JUST PLANE AWESOME

Engineer's Logbook

Build it. Wind it. Trim it. Fly it.



ENGINEER ID

Name: _____

Partner: _____

Event: Just Plane Awesome **Best flight:** _____

If found, please return to the DDMES Science Olympiad binder shelf.

How a plane engineer works

Engineers don't guess and hope. They **plan**, **test**, write down what happened, and change **one thing at a time**. A twisted rubber motor stores energy that spins a propeller. If the plane is light, strong, and trimmed to fly in a slow, gentle circle, it can stay up a long time.

1
BUILD
You build the plane at home. A grown-up keeps you safe.

2
CHECK
Weigh and measure it against the rules.

3
TRIM
Glide tests: fix stalls, dives, and tight turns first.

4
TEST
Change only the turns. Time every flight with 3 timers.

5
LOG
Write every flight with units. 10+ flights earns a bonus.

The Engineer's Code

- [ ] Safety glasses on the whole time, not on your head.
- [ ] Only students touch the plane. Grown-ups help you stay safe, but you do the building and fixing.
- [ ] Stretch-wind the motor. Hold the propeller until you launch.
- [ ] Launch gently and level, never at people.
- [ ] Write every flight down with units: seconds, turns, grams.
- [ ] Change one thing at a time so you know what worked.

My build kit

- [ ] This logbook and pencils
- [ ] Your plane kit and the plans
- [ ] Safety glasses
- [ ] A winder that counts turns
- [ ] Glue, clear tape, a little clay
- [ ] A ruler and a stopwatch

Logbook map

Know your event	p. 3-4
Plane drawings (graph paper)	p. 5-6
Trial Logs (flights)	p. 7-10
Best-flight graph	p. 11
What went wrong?	p. 12
At-home build challenges	p. 13-16
Tournament checklist	p. 17
Missions and badges	p. 18-19

Where to practice

Your kitchen table
build here (with newspaper down and a grown-up nearby)

A big indoor room
gym, church hall, or living room with no fans on

Outside on a calm day
paper gliders and trim tests

The scale at practice
weigh your plane and motors to 0.01 g

Airports and videos
watch real planes use the same 4 forces

Open Lab Fridays
the Just Plane Awesome bin has rubber bands, a stopwatch, and Flight Logs

Did you know? Indoor rubber-powered planes can fly for more than 30 minutes in giant rooms. They are so light and slow that a person could walk beside them!

Know your event: the rules

Pairs bring up to 2 student-built rubber-powered balsa planes, up to 4 motors, safety glasses, a tool kit, and an optional Flight Log (10% bonus), then get one 6-minute indoor flight period for test flights and up to 2 official flights; the longest official flight time wins.

Part	What the rules say
Bring	Up to 2 planes you built, up to 4 motors, safety glasses, a tool kit, and your Flight Log (optional, but it earns a bonus).
Flight period	6 minutes, indoors. Take test flights, then up to 2 official flights.
Say it out loud	Say TEST or OFFICIAL before each launch. Forget to say TEST, and it counts as an official flight.
Timing	3 timers on each official flight. The middle time counts.
Tool kit	Winder, glue, tape, clay, spare O-rings, sandpaper. No extra rubber.
Hands off	Only students touch the planes. Safety glasses stay on the whole time.

How it's scored

- **Score = your best official time** (in seconds).
- Flight Log passes all checks? **Multiply by 1.1** (10 percent more).
- Ties go to the higher **second** official flight.
- Any broken building rule puts you in **Tier 2**, below every team with no broken rules.

Score math

Best official flight: 40 seconds
Flight Log passes: x 1.1
Score = 44 seconds

Log has only 8 flights? No bonus. It needs 10 or more.

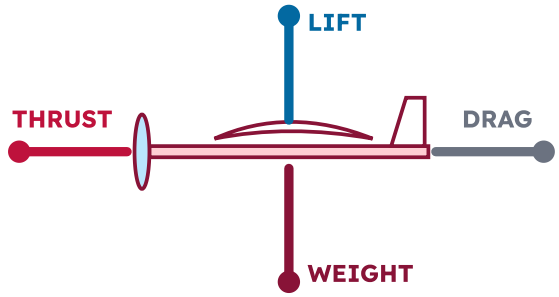
Plane rule check (from Lesson 2)

Rule	Limit	My plane	OK?
Mass without motor	7.5 g or more		☐ OK
Wingspan	41.0 cm or less		☐ OK
Propeller	One piece plastic, 15.5 cm or less		☐ OK
Each motor (with O-rings)	2.0 g or less		☐ OK
Made of	Wood, glue, tissue (no foam)		☐ OK
Label	Team or school name on plane		☐ OK

Pass all 6 checks to earn the **Rule Checker** badge.

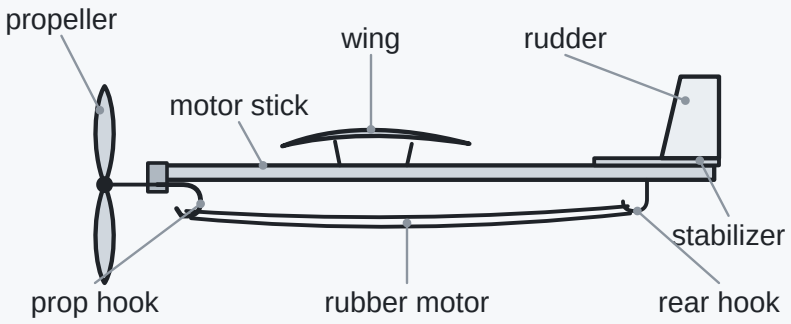
Know your science: how it flies

The four forces



Lift	The upward push from air on the wing.
Weight	Gravity pulling the plane down.
Thrust	The forward pull from the spinning propeller.
Drag	Air pushing back on the plane and slowing it down.

The 8 plane parts



Energy words

Stored energy	Energy waiting to be used. A twisted rubber band stores energy.
Energy of motion	Energy something has when it moves. The unwinding rubber spins your card (or propeller).
Heat	Some energy always turns into a little heat. Stretch a rubber band fast and touch it to your lip: it feels warm.
Fair test	Change only one thing. Keep everything else the same.

How to stretch-wind

1. Glasses on.
2. Partner holds the plane at the back of the motor stick.
3. Hook the winder to the front of the motor. Stretch the motor to about 3 times its length.
4. Wind and count. Walk slowly back toward the plane as you finish.
5. Hold the propeller, move the motor onto the prop hook, and launch gently and level.

Energy path: your arm winds the motor → **stored energy** in the twisted rubber → the propeller spins (**energy of motion**) → **thrust** pulls the plane forward → the wing makes **lift**. A little energy always turns into heat.

Flight Log bonus check (all needed for the x 1.1 bonus)

- ☐ Uses the official Flight Log page from the Rules Manual
- ☐ 10 or more test flights recorded
- ☐ Column 2 (flight time, seconds) filled in for every flight
- ☐ Column 3 (turns of winder) filled in for every flight
- ☐ At least one more column filled in: motor mass (grams), motor width (inches), or ceiling height (feet)
- ☐ Units written in the headers
- ☐ School name and our names at the top

My plane: top view

Draw your plane life size if it fits, or use the scale below. Label all 8 parts: wing, motor stick, stabilizer, rudder, propeller, prop hook, rear hook, rubber motor. Mark every repair and trim change with the date.

Design name _____ **Date** _____ **Version #** _____

Top view

1 small square = ____ cm

Wingspan (41.0 cm max) and prop (15.5 cm max):

Mass without motor (7.5 g or more):

My plane: side view and trim

Draw your plane life size if it fits, or use the scale below. Label all 8 parts: wing, motor stick, stabilizer, rudder, propeller, prop hook, rear hook, rubber motor. Mark every repair and trim change with the date.

Design name _____ **Date** _____ **Version #** _____

Side view

1 small square = ____ cm

Where my nose clay is and how much:

My best turn count so far and why:

Trial Log #1

Every flight is a trial. Result: turns wound and flight time in seconds (middle of 3 timers). Copy every flight to your official Flight Log with units: time (seconds), turns, and motor mass (grams). 10+ flights earns the 10 percent bonus.

Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My prediction (seconds)	Result (turns, seconds)	What I learned
1				
2				
3				
4				
5				
6				
7				
8				

Best trial on this page: # _____ **Why it worked:** _____

Trial Log #2

Every flight is a trial. Result: turns wound and flight time in seconds (middle of 3 timers). Copy every flight to your official Flight Log with units: time (seconds), turns, and motor mass (grams). 10+ flights earns the 10 percent bonus.

Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My prediction (seconds)	Result (turns, seconds)	What I learned
9				
10				
11				
12				
13				
14				
15				
16				

Best trial on this page: # _____ **Why it worked:** _____

Trial Log #3

Every flight is a trial. Result: turns wound and flight time in seconds (middle of 3 timers). Copy every flight to your official Flight Log with units: time (seconds), turns, and motor mass (grams). 10+ flights earns the 10 percent bonus.

Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My prediction (seconds)	Result (turns, seconds)	What I learned
17				
18				
19				
20				
21				
22				
23				
24				

Best trial on this page: # _____ **Why it worked:** _____

Trial Log #4

Every flight is a trial. Result: turns wound and flight time in seconds (middle of 3 timers). Copy every flight to your official Flight Log with units: time (seconds), turns, and motor mass (grams). 10+ flights earns the 10 percent bonus.

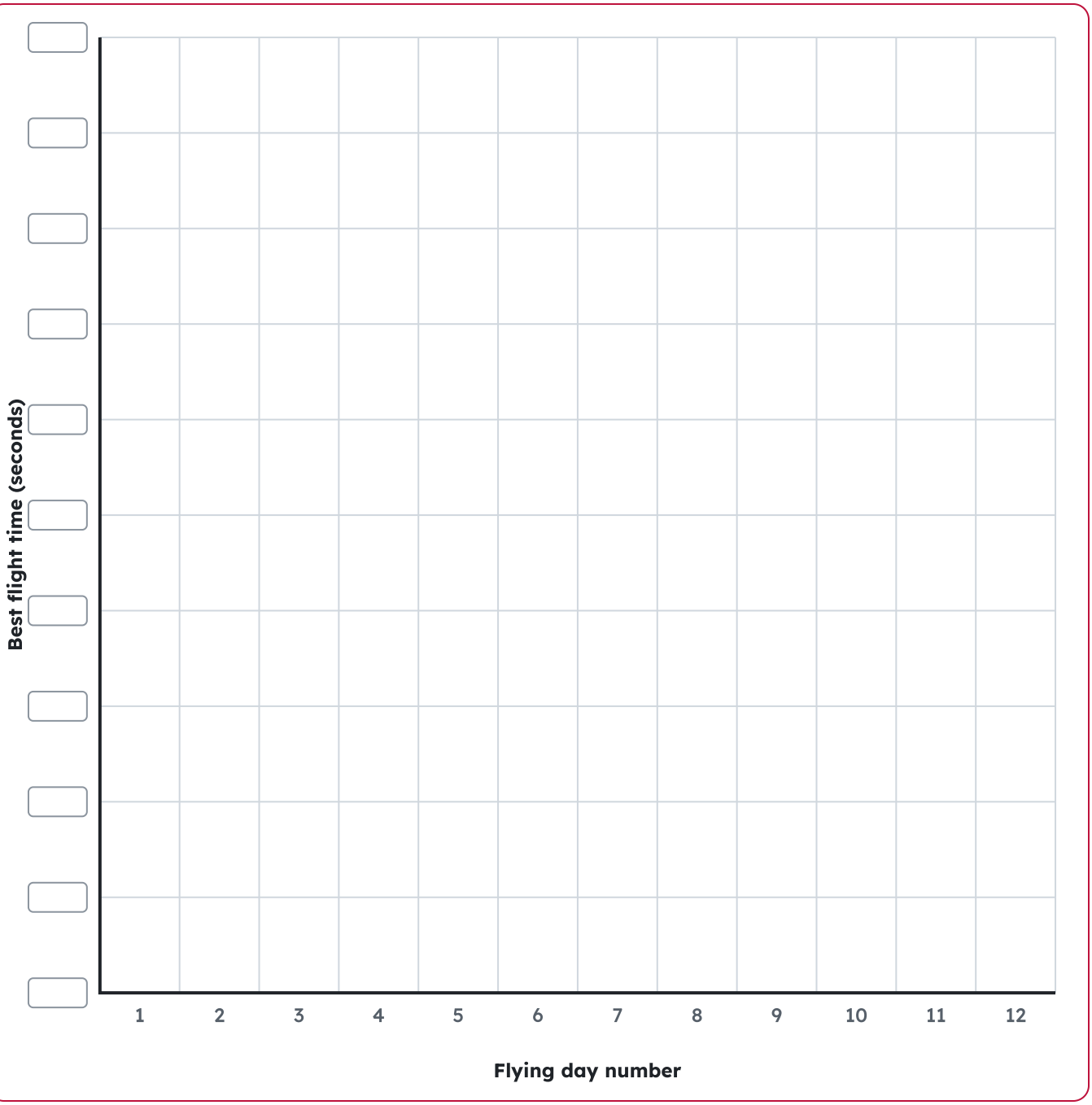
Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My prediction (seconds)	Result (turns, seconds)	What I learned
25				
26				
27				
28				
29				
30				
31				
32				

Best trial on this page: # _____ **Why it worked:** _____

My best-flight graph

After each flying day, put a dot for your longest flight (the middle of 3 timers). Write the numbers on the side (count by 5s or 10s), then connect the dots.



Date of #	1	2	3	4	5	6	7	8	9	10	11	12

Higher is better! Next to each dot, write how many turns you wound. Do more turns give longer flights? When does the plane start hitting the ceiling?

My best so far: _____

My tournament goal: _____

Is my line going the right way? _____

What went wrong?

Every engineer's plane crashes sometimes. Good engineers figure out **why**. Fix the trim first, then add turns. Winding more turns makes a stall worse!

Problem	What it looks like	Try this (one fix at a time)
Stall	Nose goes up, plane slows down, then drops	Add a tiny bit of clay to the nose, or slide the wing back a little
Dive	Nose points down and plane hits the floor fast	Remove some nose clay, or slide the wing forward a little
Turn too tight	Small circles, plane spirals down	Bend the rudder a little less
Flies straight into walls	No circle	Bend the rudder a little toward the side you want it to circle
Climbs too steeply at the start	Shoots up, may hit ceiling	Wind fewer turns until the trim is right
Motor wastes energy	The rubber bunches up and the prop stops early.	Stretch the motor to about 3 times its length while you wind. Walk back as you finish.
Something broke	A cracked stick, a loose hook, torn tissue.	You fix it: a small dot of glue, a bit of tape. Let glue dry before you fly.
Ran out of time	The 6 minutes end before an official flight.	Plan an official flight by minute 3. Say TEST or OFFICIAL every time.

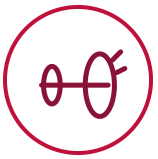
My fix-it log

Write down each problem you hit. Fix 3 problems to earn the **Crash Fixer** badge.

What went wrong	Why I think it happened	What I tried	Did it work?
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no

AT-HOME BUILD CHALLENGE 1 · AFTER LESSON 1 (FRI OCT 23)

Straw Hoop Glider



Glider Maker badge

Your mission: build a glider from a straw and 2 paper hoops. Make it glide smoothly, not stall or dive, just like trimming your real plane.

You need

- [ ] A plastic straw
- [ ] An index card or stiff paper
- [ ] Tape and scissors
- [ ] A ruler
- [ ] 1 or 2 paper clips (for nose weight)

Build it

1. Cut 2 strips 2.5 cm wide: one 13 cm long, one 25 cm long.
2. Tape each strip into a hoop.
3. Tape the small hoop to the front of the straw and the big hoop to the back.
4. Hold the middle and push it forward gently and level (don't throw it like a ball). Time it.
5. Stall or dive? Try a paper clip on the front, then move it. Change one thing at a time.

Draw your glider and mark where the weight worked best

My results

Try	What I changed	Time (sec)	Stall, dive, or smooth?
1			
2			
3			

Which of the four forces did your push make? Which force slowed the glider down?

Safety: glasses on, and launch away from people and pets.

Date done: _____ **Coach or Captain initials (Glider Maker badge):** _____

AT-HOME BUILD CHALLENGE 2 · AFTER LESSON 2 (FRI DEC 4)

Rubber-Band Roller



Energy Winder badge

Your mission: build a roller powered by a twisted rubber band. Find out how the number of turns changes how far it goes, like the turns column in your Flight Log.

You need

- [ ] An empty thread spool
- [ ] A rubber band a little longer than the spool
- [ ] A pencil and a toothpick
- [ ] Tape and a paper clip
- [ ] A tape measure or meter stick

Build it

1. Poke the rubber band through the spool hole with the paper clip.
2. Hold one end with a short piece of toothpick taped to the spool.
3. Put the pencil through the loop on the other end.
4. Wind 10 turns, set it on the floor, and let go. Measure how far it rolled.
5. Try 20 and 30 turns. Do each count twice. Change ONLY the turns.

Draw your roller, or graph turns (across) vs distance (up)

My results

Turns	Distance, try 1 (cm)	Distance, try 2 (cm)	Best (cm)
10			
20			
30			

Where was the energy stored? What did it turn into? What happened when you added turns?

Safety: keep your face away from the twisted rubber band. Let go gently.

Date done: _____ **Coach or Captain initials (Energy Winder badge):** _____

AT-HOME BUILD CHALLENGE 3 · AFTER LESSON 3 (FRI JAN 22)

Paper Plane Trim Lab

Trim Tuner badge

Your mission: fold a paper airplane, then **trim** it: fix a stall, a dive, and a turn, one small change at a time.

You need

- ☐ Printer paper
- ☐ 1 or 2 paper clips (nose weight, like clay)
- ☐ A timer
- ☐ A big room or a calm day outside

Build it

1. Fold your best paper airplane. Fly it 2 times gently and level.
2. Stall (nose up, then drops)? Add a paper clip to the nose.
3. Dive (nose down fast)? Move the clip back, or bend the back edges up a tiny bit.
4. Turning? Bend the tail edge a tiny bit toward the side you want it to turn.
5. Log each flight. One change at a time!

Draw your plane and mark each trim change

My results

Flight	What I changed (one thing)	Stall, dive, turn, or smooth?	Time (sec)
1			
2			
3			
4			

Why should you change only one thing at a time?

Safety: never throw planes at people or pets.

Date done: _____ **Coach or Captain initials (Trim Tuner badge):** _____

AT-HOME BUILD CHALLENGE 4 · AFTER LESSON 4 (FRI FEB 19)

Hangar Box

Hangar Builder badge

Your mission: build a carry box that keeps 2 planes safe on tournament day, with a spot for motors, the winder, and your tool kit.

You need

- [ ] A shoebox (or a bigger box if your wings are long)
- [ ] Cardboard scraps
- [ ] Tape and scissors
- [ ] A marker
- [ ] A zip bag for motors

Build it

1. Measure your wingspan. Will the plane fit? If not, find a longer box.
2. Tape cardboard strips inside as racks so the wings don't touch the sides.
3. Make a pocket for motors (in a zip bag), the winder, and the tool kit.
4. Label the box with your team or school name.
5. Test it: put a plane in, carry the box around the house. Did anything move?

Draw your hangar box plan with measurements

My results

Check	Yes / no	Fix I made
2 planes fit, nothing touching		
Motors, winder, tool kit have a spot		
Survived the carry test		

What was the hardest part to protect? How did you solve it?

Safety: an adult helps with cutting. Only you handle your planes.

Date done: _____ **Coach or Captain initials (Hangar Builder badge):** _____

Tournament-day checklist

Sat Mar 13, Chatham Regional (Seaforth High School). Check each box as you go.

Pack your bag

- ☐ Up to 2 planes, in your hangar box
- ☐ Up to 4 motors (each 2.0 g or less)
- ☐ Winder
- ☐ Tool kit: glue, tape, clay, spare O-rings, sandpaper (NO extra rubber)
- ☐ Official Flight Log (10+ flights, with units)
- ☐ Safety glasses

The night before

- ☐ Check both planes against the 6 rules (p. 3).
- ☐ Finish your repair list. You do the work.
- ☐ Decide your turn count from your graph.
- ☐ Get a good night's sleep.

Rules to remember

- 6-minute flight period.
- Up to 2 official flights; the best one counts.
- Only students touch the planes.
- No extra rubber in the tool kit.

At the event

- ☐ Glasses on before you walk in.
- ☐ Check-in: the judges weigh and measure your plane.
- ☐ Test flight at your best turn count. Say TEST.
- ☐ Fix trim if needed (one change). Official flight 1 by minute 3. Say OFFICIAL.
- ☐ Official flight 2 (second plane or a re-wound motor). Say OFFICIAL.

After: my tournament report

Our score or time:

What went great:

What we would change:

Thank you to:

Draw or write my favorite moment of the day

Build Missions

One mission for each Just Plane Awesome meeting. When you finish, show a coach or your Team Captain for initials, then color in the badge on page 19 and in your Team Badge Passport.

When	Mission	Badge	Done	Initials
After Fri Oct 23 Lesson 1	Build and trim a Straw Hoop Glider (p. 13).	Glider Maker	☐	
After Fri Dec 4 Lesson 2	Rubber-Band Roller: test 10, 20, and 30 turns (p. 14).	Energy Winder	☐	
After Fri Jan 22 Lesson 3	Paper Plane Trim Lab: fix a stall, a dive, and a turn (p. 15).	Trim Tuner	☐	
After Fri Feb 19 Lesson 4	Build a Hangar Box for tournament day (p. 16).	Hangar Builder	☐	
Anytime Build	Build your plane yourself and draw it (p. 5-6). Pass all 6 rule checks (p. 3).	Build Boss, Rule Checker	☐	
Anytime Log	Log 10+ flights (p. 7-10) and graph your best (p. 11).	Flight Logger, Graph Maker	☐	
Anytime Bonus	Fix 3 problems on What went wrong? (p. 12).	Crash Fixer	☐	

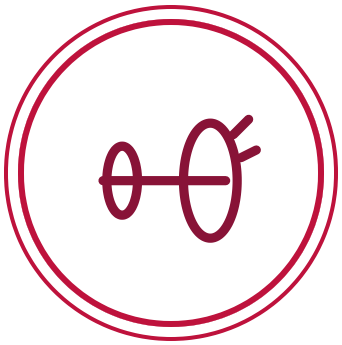
Tournament day: Sat Mar 13, Chatham Regional (Seaforth High School)

- ☐ Planes, motors, winder
- ☐ Tool kit and Flight Log
- ☐ Safety glasses

- Full checklist on page 17.
- Trim first, then turns.
- Official flight by minute 3.

My Badges

Color in each badge when you earn it. Earn all 9 to become a **Master Plane Engineer**! Your Team Badge Passport has the same badges.



Glider Maker

Build and trim a straw hoop glider

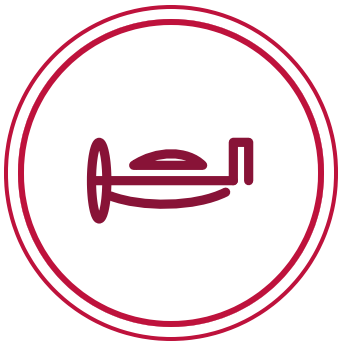
Initials: _____



Energy Winder

Rubber-band roller: turns vs distance

Initials: _____



Trim Tuner

Fix a stall, a dive, and a turn

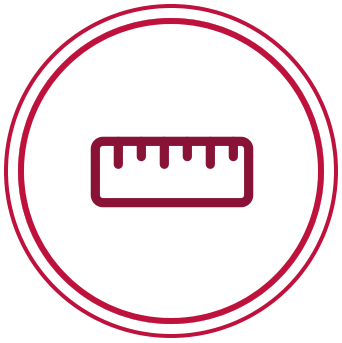
Initials: _____



Hangar Builder

Build a carry box for your planes

Initials: _____



Rule Checker

Your plane passes all 6 rule checks

Initials: _____



Flight Logger

10+ flights logged, with units

Initials: _____



Graph Maker

Graph your best flight times

Initials: _____



Crash Fixer

Fix 3 problems on What went wrong?

Initials: _____



Build Boss

Build your plane yourself and draw it

Initials: _____